
CAIRO Regression: Calibrate After Initial Rank Ordering

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Abstract

Many regression tasks care about both the *ordering* of predictions and their *scale*. We propose CAIRO (Calibrate After Initial Rank Ordering), a simple two-stage framework for real-valued regression. Stage 1 trains a score function using rank-based losses; Stage 2 fits a one-dimensional monotone regression on a held-out calibration set to map scores back to the scale of the targets. Empirically, CAIRO acts as a robust regression method: it effectively shields against heavy-tailed outliers, stabilizing neural training and matching the performance of tree ensembles on tabular benchmarks.

1. Introduction

Supervised learning with continuous targets is traditionally dominated by two distinct paradigms. On one side is *regression*, which prioritizes the numerical scale of predictions, typically by minimizing pointwise distances such as Mean Squared Error (MSE). On the other side is *learning-to-rank*, which prioritizes the relative ordering of instances, optimizing pairwise or listwise objectives while disregarding the absolute scale of the output. In many high-stakes applications—such as insurance pricing, credit scoring, and personalized medicine—this dichotomy is artificial and limiting. A robust risk model must not only rank high-risk individuals correctly (ordering) but also predict their expected costs accurately (calibration).

Standard regression models often fail to capture the correct ordering in complex, heavy-tailed regimes, as they over-focus on fitting the conditional mean to minimize large residuals. Conversely, ranking models produce score distributions that are scale-invariant and thus unusable for downstream tasks requiring valid units, such as dollar amounts or probabilities.

We introduce CAIRO (Calibrate After **I**nitial **R**ank **O**rdering), a two-stage framework that treats regression as a composite task: learning the order of the data through ranking algorithms, followed by its scale with recalibration methods. This separation allows CAIRO to function as a form of robust regression: it utilizes the scale-invariance of ranking to ignore the magnitude of outliers during representation learning, reintroducing scale only via a constrained calibration step

The remainder of this paper is organized as follows. Section 2 reviews state-of-the-art approaches in learning-to-rank and calibration mechanisms, discussing previous attempts to combine these objectives. In Section 3, we formalize the CAIRO methodology. In Section 4, we rigorously define the class of loss functions eligible to be used in Stage 1 and introduce smooth empirical loss functions suitable for gradient-based training. We further show in Section 5 that calibrating a ranker trained on these objectives guarantees the recovery of the true conditional mean. Finally, Section 6 presents a comprehensive data analysis on both synthetic and real-world benchmarks. We demonstrate that CAIRO offers clear advantages in heteroskedastic and heavy-tailed noise regimes—settings where standard MSE minimization struggles—often matching or exceeding strong tree-based baselines.

2. Background and Related Work

2.1. Regression versus Ranking

In classical regression, the goal is to estimate the conditional mean $m^*(x) = \mathbb{E}[Y \mid X = x]$ by minimizing a pointwise risk such as $\mathbb{E}[(Y - f(X))^2]$. This viewpoint is scale-aware but agnostic to how often the induced scores $f(X)$ correctly order pairs (X, X') according to Y .

Ranking formulations adopt the opposite stance. Given two i.i.d. copies (X, Y) and (X', Y') , a score g is evaluated by how often it preserves the sign of label differences, e.g. through the pairwise misranking probability

$$\mathbb{E}[\mathbb{1}\{(Y - Y') \text{ sign}(g(X) - g(X')) \leq 0\}], \quad (1)$$

Such objectives are inherently *scale-free*. This tension between accurate *scale* and accurate *order* motivates methods that can explicitly account for both.

2.2. Learning to Rank: Pairwise and Listwise Surrogates

The learning-to-rank literature, especially in information retrieval, has developed a mature toolbox of ranking algorithms. Pairwise methods such as RankSVM (Joachims, 2002) and RankNet / LambdaRank / LambdaMART (Burges et al., 2005; Burges, 2010) optimize smooth surrogates of pairwise misordering—typically logistic or hinge-style losses applied to score differences $g(X) - g(X')$, often augmented with heuristic importance weights that reflect ranking metrics like NDCG or MAP (?). Listwise methods (e.g., ListNet, ListMLE, SoftRank) instead construct losses at the level of entire ranked lists, sometimes via differentiable relaxations of the sorting operator (Cao et al., 2007; Taylor et al., 2008). More recent work introduces permutation-based relaxations such as *SoftSort*, which approximate ranks and sorted scores with smooth operators suitable for gradient-based training (Prillo & Eisenschlos, 2020).

These methods directly target ordering quality, but they almost never aim to recover m^* or produce calibrated real-valued predictions. Scores are typically evaluated via rank metrics rather than squared error or other regression losses, and theoretical guarantees are cast in terms of surrogate consistency for ranking risks. Moreover, many learning-to-rank setups work with discrete relevance labels rather than continuous responses. As a result, they offer limited guidance for how to use powerful rank-based objectives to ultimately obtain a calibrated regression model.

2.3. Recalibration of Scores

Calibration addresses the question of how to map scores back onto the scale of observed outcomes. In binary classification, post-hoc calibration methods such as Platt scaling, isotonic regression, and related techniques are widely used to convert arbitrary scores into calibrated probabilities (Platt, 1999; Zadrozny & Elkan, 2002; Niculescu-Mizil & Caruana, 2005).

More broadly, shape-constrained regression offers a range of monotone regression and calibration tools, including isotonic least squares and monotone splines (Barlow et al., 1972; Robertson et al., 1988; Meyer, 2008; Guntuboyina & Sen, 2018). These approaches share a common structure: they solve a convex (or nearly convex) optimization problem in one dimension (the score axis) under a monotonicity constraint, and they are often used as a lightweight post-processing step on top of a complex base model.

Existing calibration work, however, typically treats the base score function as given. The score may come from a regression model, a classifier, or a ranking method, but the calibration procedure is agnostic to *how* that score was trained and to whether its population target is related in a simple way to m^* .

2.4. Hybrid and Multi-Stage Approaches

Several works have attempted to combine ranking and regression goals more directly. Hybrid objectives that additively combine pointwise regression losses with pairwise ranking surrogates have been proposed to trade off ordering quality and numerical accuracy within a single model (Sculley, 2010). In the context of deep neural rankers, Yan et al. (2022) study *scale calibration* of deep ranking models and empirically show that naively mixing regression and ranking losses can lead to conflicting gradients and poorly calibrated scores, motivating specialized objectives that are explicitly designed to be regression-compatible. These approaches share the intuition that ranking and scale should be aligned, but they are developed for retrieval-style relevance settings rather than general real-valued regression.

Sequential approaches: rank then calibrate. A cleaner alternative, introduced by Menon et al. (2012), is to decouple the two tasks. A first-stage model learns to order examples using a ranking-oriented loss, and a second-stage regression then calibrates the resulting scores. However, existing formulations often rely on restrictive assumptions such as linearity, binary outcomes, or limited calibration families, reducing their generality in modern machine learning settings.

More recent work extends this two-step principle to continuous targets. Wüthrich & Ziegel (2023) study isotonic recalibration of regression scores, emphasizing that success depends on the initial ranker preserving the ordering of conditional means. If that ordering is lost—e.g., under low signal-to-noise—monotone post-processing cannot recover it. This observation motivates methods that *prioritize high-quality ranking* before calibration.

2.5. Motivation for a Rank–Then–Calibrate Framework

The picture that emerges from these threads is the following:

- Learning-to-rank provides powerful pairwise and listwise losses, but largely ignores the downstream goal of calibrated regression.
- Monotone recalibration methods can restore scale given a good score, but are typically applied as a generic post-hoc tool without tailoring the training objective of the score itself.

This suggests a simple but underexplored strategy for regression with real-valued targets: *first* learn a score function whose population minimizers are precisely monotone transformations of m^* , and *then* apply a one-dimensional monotone recalibration step to recover the scale of m^* .

3. Our Method

We propose a two-stage population learning framework to construct a regression model. In Stage 1, a model is trained using a rank-based loss $\mathcal{L}_{\text{rank}}$ over a large function class $\mathcal{H}$ to learn the ordering structure of m^* , where $m^*(x) = \mathbb{E}[Y|X = x]$ is the true regression function.

In Stage 2, the Stage 1 output is calibrated with isotonic regression. We choose isotonic regression in Stage 2 since it minimizes the Least Squares loss $\mathcal{L}_{\text{iso}}$ over the space of non-decreasing functions $\mathcal{T}$, which preserves the ranking learned in Stage 1. Formally,

$$\begin{aligned} g &\in \arg \min_{h \in \mathcal{H}} \mathcal{L}_{\text{rank}}(Y, h(X)), \\ f &\in \arg \min_{t \in \mathcal{T}} \mathcal{L}_{\text{iso}}(Y, t(g(X))). \end{aligned}$$

Later, we will show that we expect the optimal model out of stage 1 to be of the form $g(X) = h(m^*(X))$ for h strictly increasing. Hence, the isotonic regression step can be viewed as learning the inverse of this transformation, thereby recovering m^* in its original scale. From here on out, we refer to this two-stage framework as the Calibrated After Initial Rank Ordering (CAIRO) framework.

4. Ranking Phase

4.1. Population Losses for Ranking

The CAIRO framework does not impose a single, fixed ranking loss. Instead, it accommodates an entire family of rank-based objectives. We call such losses *Optimal in Rank Order* (ORO) and define them below.

Definition 4.1. A loss functional $\mathcal{L}(g)$ is Optimal in Rank Order (ORO) if every population minimizer belongs to

$$\mathcal{H}' = \{\phi \circ m^* : \phi : \mathbb{R} \rightarrow \mathbb{R} \text{ strictly increasing}\} \subset \mathcal{H}.$$

Examples of ORO losses. The first example we give of an ORO loss is the Pairwise loss presented in (1). The fact that the pairwise loss is ORO is shown in (Cl  men  on et al., 2008). Moreover, we can view the Pairwise loss as a generator of ORO losses. In particular, we show that a weighted variant of the pairwise loss is ORO.

Proposition 4.2 (Weighted pairwise ORO losses). *Let (X, Y) and (X', Y') be i.i.d. pairs. Consider the weighted pairwise misranking loss*

$$\mathcal{L}_w(g) = \mathbb{E}[|Y - Y'| \mathbb{1}\{(Y - Y') \text{ sign}(g(X) - g(X')) \leq 0\}], \quad (2)$$

Then the loss $\mathcal{L}_w$ is ORO. The proof is deferred to Appendix A.3.

Weighted variants of the pairwise loss do not fully characterize the entire class of ORO functionals, but they include the specific objectives we study in our numerical implementation. Beyond guaranteeing that these objectives target the correct rank order, the absolute-gap variant in particular admits a clear connection to a well-known measure of association.

Proposition 4.3. *Under the assumptions of Proposition 4.2, let $F_{g(X)}$ denote the CDF of $g(X)$. Then:*

$$\mathcal{L}_w(g) = \frac{1}{2} \mathbb{E}[|Y - Y'|] - 2 \text{Cov}(Y, F_{g(X)}(g(X))), \quad (3)$$

i.e., the loss differs from a Gini-type covariance between Y and $g(X)$ only by an additive constant. This identity is proved in Proposition A.4 in the Appendix.

Absolute-gap weights (Gini-style loss). With $w = |Y - Y'|$, pairs whose labels are far apart receive larger weight. Equation (3) shows that the absolute-gap weighted loss is, up to fixed constants, the negative of a Gini-type covariance between the response Y and the model score distribution $F_{g(X)}(g(X))$. This quantity is defined as the *Gini covariance* in the income-inequality and actuarial literature, where it is used as a measure of association that is sensitive to the scale of Y while depending on g only through its induced ranks.

The Gini covariance has been extensively studied in actuarial science. Recently, Wüthrich & Ziegel (2024) examined it as a loss function and concluded that it is only sensible to optimize it over the class of auto-calibrated functions—a strict condition that is impractical for most deep learning models, which typically operate over much larger function classes. This observation lends further credibility to CAIRO, which imposes no such restriction on the function class. The relaxation effectively removes the restrictive auto-calibration requirement in Stage 1, making the approach compatible with broad modern machine learning architectures.

4.2. Empirical Losses for Ranking

The population losses in Proposition 4.2 are not directly observable, so we approximate them with a unified empirical risk that mirrors their pairwise structure. In doing so, we connect our constructions to standard learning-to-rank objectives. Let $\{(x_i, y_i)\}_{i=1}^n$ be an i.i.d. sample, and assume for simplicity there are no ties in either the labels y_i or the scores $g(x_i)$.

Given a symmetric pairwise weight matrix $w_{ij} = w_{ji} \geq 0$, we define the empirical misranking loss

$$\ell_w(g) := \frac{1}{n(n-1)} \sum_{1 \leq i < j \leq n} w_{ij} \mathbb{1}\{(y_i - y_j)(g(x_i) - g(x_j)) \leq 0\}. \quad (4)$$

This single template covers all of our Stage 1 objectives; different choices of w_{ij} correspond to the different population ORO losses introduced earlier.

Under the no-ties assumption on $\{y_i\}$ and $\{g(x_i)\}$, $\ell_w(g)$ also admits the ordered-pair form

$$\ell_w(g) = \frac{1}{n(n-1)} \sum_{i \neq j} w_{ij} \mathbb{1}\{y_i > y_j\} \mathbb{1}\{g(x_i) < g(x_j)\}, \quad (5)$$

whose derivation is given in Appendix A.4. In practice we use two plug-in schemes that mirror the population weights: $w_{ij}^{(0)} := 1$ and $w_{ij}^{(G)} := |y_i - y_j|$.

4.3. Smooth Surrogates for Empirical Losses

Finally, because these empirical objectives are non-differentiable, we replace the hard indicator in (5) with smooth surrogates.

Method 1: Log-Sigmoid Upper Bound. Classical treatments have long used the log-sigmoid as a smooth, convex surrogate for the indicator function. In particular, this choice is justified by the following upper bound on the indicator:

$$\mathbb{1}\{g(x_i) < g(x_j)\} \leq \log_2 \left(1 + e^{-\sigma(g(x_i) - g(x_j))} \right), \quad (6)$$

The σ is a hyperparameter that characterizes the shape of the sigmoid. It then follows that we can use this relationship to derive an upper bound on the empirical estimator of our loss function as:

$$\ell_w(g) \leq \frac{1}{n(n-1)} \sum_{i \neq j} w_{ij} \mathbb{1}\{y_i > y_j\} \log_2 \left(1 + e^{-\sigma(g(x_i) - g(x_j))} \right). \quad (7)$$

Example 1. Consider the Pairwise loss Estimator with $w_{ij} = 1$. Equation (7) reduces to

$$\ell_w(g) \leq \frac{1}{n(n-1)} \sum_{i \neq j} \mathbb{1}\{y_i > y_j\} \log_2 \left(1 + e^{-\sigma(g(x_i) - g(x_j))} \right).$$

We recognize the upper bound to be equivalent to the RankNet loss. Thus RankNet appears as the uniform-weight special case of our general weighted pairwise framework.

This approach has demonstrated strong empirical performance in producing effective ranking algorithms. Nevertheless, we also examine an alternative smoothing strategy.

Method 2: Smooth Rank Surrogate. Rather than smoothing the *pairwise indicator*, we first rewrite the statistic in terms of a *rank* operator and then replace $\text{rank}(\cdot)$ by a differentiable, permutation-based relaxation (e.g., *SoftSort*; Prillo & Eisenschlos, 2020). This yields a listwise objective with global, position-aware gradients.

Example 2. Consider the Pairwise loss estimator with $w_{ij} = |y_i - y_j|$. Using (5), we obtain

$$\ell_w(g) = \frac{1}{n(n-1)} \sum_{i \neq j} |y_i - y_j| \mathbb{1}\{y_i > y_j\} \mathbb{1}\{g(x_i) < g(x_j)\} \quad (8)$$

$$\equiv \sum_{i=1}^n y_i \text{rank}(g(x_i)) \quad (\text{up to additive constants}); \text{ see (Chen et al., 2019).} \quad (9)$$

We obtain a smooth, ML-ready objective by substituting the *SoftSort* soft rank:

$$\ell_w(g) := \sum_{i=1}^n y_i \text{softrank}(g(x_i)), \quad (10)$$

which directly relaxes (8) without resorting to a pairwise upper bound. In practice this is $O(n \log n)$ per batch (sort + linear-time operations), and it aligns training with the intended listwise target (Gini/ARP) while providing smooth gradients end-to-end.

Remark. The rank-based form of Eq. 8 has a direct connection to the empirical Gini index. Frees et al. (2011) proposed a consistent estimator for the Gini index that is, up to positive scaling constants, exactly equivalent to this quantity—an equivalence that naturally follows from our population-level motivation.

Both smoothing paradigms—pairwise log-sigmoid bounds and listwise soft-rank relaxations—have complementary strengths: the pairwise loss is simple and computationally light, while the soft-rank formulation more directly targets the desired rank metric. In our experiments we compare them for the gap-weighted (Gini-style) objective and observe the expected trade-offs between bias, smoothness, and cost.

5. Calibration Stage

CAIRO is a two-stage framework: Stage 1 learns an *ordering* that is correct up to a strictly increasing transform; Stage 2 restores the *scale* of the response. Let $m^*(x) = \mathbb{E}[Y \mid X = x]$ and let $g : \mathcal{X} \rightarrow \mathbb{R}$ denote the Stage 1 score. Under any ORO loss, an optimal population ranker satisfies $g(x) = h \circ m^*(x)$ for some strictly increasing $h : \mathbb{R} \rightarrow \mathbb{R}$. The goal of Stage 2 is therefore not to modify this ordering, but to learn a monotone map that inverts the unknown transform h and recovers m^* on the original scale of the response.

5.1. Population Calibration Problem

Given scores $S = g(X)$, the population calibration problem seeks a nondecreasing function that best aligns these scores with the outcome Y . Let $\mathcal{M} := \{t : \mathbb{R} \rightarrow \mathbb{R} \text{ nondecreasing}\}$. We define the population objective

$$f \in \arg \min_{t \in \mathcal{M}} \mathbb{E}[(Y - t(S))^2]. \quad (11)$$

If the Stage 1 score is of the form $g = h \circ m^*$ for a strictly increasing link h , then the ordering of the scores reflects the ordering of the true regression function m^* . In this setting the optimizer of (11) agrees with m^* when composed with g , so calibration recovers the correct regression function.

The following theorem formalizes this statement; the proof is given in Appendix A.2.

Theorem 5.1 (Stagewise recovery of m^*). *Assume $\mathbb{E}[Y^2] < \infty$ and let $m^*(x) = \mathbb{E}[Y \mid X = x]$. Suppose the Stage 1 score satisfies $g(x) = h \circ m^*(x)$ for some strictly increasing $h : \mathbb{R} \rightarrow \mathbb{R}$.*

Let f be any minimizer of the monotone least-squares problem (11). Then

$$f(g(X)) = m^*(X) \quad \text{a.s.}$$

5.2. Empirical Isotonic Regression

In practice we observe $\{(x_i, y_i)\}_{i=1}^n$ and compute Stage 1 scores $s_i := g(x_i)$. The empirical analogue of (11) is

$$\min_{\tilde{t}} \frac{1}{n} \sum_{i=1}^n (y_i - \tilde{t}(s_i))^2 \quad \text{s.t.} \quad \tilde{t}(s_i) \leq \tilde{t}(s_j) \text{ whenever } s_i \leq s_j. \quad (12)$$

Writing the scores in nondecreasing order $s_{(1)} \leq \dots \leq s_{(n)}$ and reindexing $y_{(i)}$ accordingly, the classical Pooled Adjacent Violators (PAV) algorithm (Barlow et al., 1972) computes the unique minimizer of (12) in $O(n)$ time after sorting, so the total cost per calibration fit is $O(n \log n)$.

Let $\hat{t}_{\text{iso}}$ denote the resulting isotonic fit on the score axis, defined at the observed scores $s_{(i)}$ and extended piecewise-constantly between them. For prediction we then use

$$\hat{m}_{\text{CAIRO}}(x) := \hat{t}_{\text{iso}}(g(x)),$$

with standard choices such as clipping scores outside $[s_{(1)}, s_{(n)}]$ or linearly interpolating between adjacent fitted values.

Under mild regularity conditions (e.g., bounded second moments and a nondegenerate distribution of S), the isotonic least-squares estimator $\hat{t}_{\text{iso}}$ is consistent for f in (11); see, for instance, Robertson et al. (1988); Guntuboyina & Sen (2018) for detailed statements. Combined with an ORO-consistent Stage 1, this implies that $\hat{m}_{\text{CAIRO}}(X)$ converges to $m^*(X)$ in L^2 , aligning the empirical procedure with the population result of Theorem A.2. In our implementation, Stage 2 is applied either on a held-out calibration set.

Computational considerations. All of the above methods retain the central structural advantage of Stage 2: the calibration problem is one-dimensional in $S = g(X)$ and involves convex (or at least quasiconvex) optimization. Monotone splines and nearly isotonic regression can be solved with specialized algorithms or standard quadratic programming, with per-iteration cost $O(n)$ – $O(n \log n)$ comparable to PAV. Bregman-based monotone regression often admits similar algorithms after suitable reparameterization. In all cases, the calibration step remains modest in cost compared to training the Stage 1 ranker, and can be treated as a lightweight post-processing stage.

6. Numerical Studies

We now empirically investigate the proposed CAIRO framework using a combination of controlled simulations and a real-world regression task. Our goals are (i) to compare regression accuracy (RMSE) and ranking quality across several noise regimes, and (ii) to assess whether the advantages of rank–then–calibrate training transfer to a practical application (Experiment 2).

Stage 1 models and architecture. We consider three Stage 1 ranking losses: *RankNet*, a pairwise logistic loss with uniform weights; *RankNet–GiniW*, the same logistic loss with absolute-gap weights $|y_i - y_j|$; We implement Equation (7) with the specified weights. We consider a third model: *GiniNet–SoftRank*, a listwise SoftRank-style surrogate for the normalized Gini coefficient which we implement with Equation (10). All three use the same two-layer MLP with hidden sizes (32, 16); the RankNet variants employ LeakyReLU activations, while GiniNet–SoftRank uses ReLU. Stage 2 applies empirical isotonic regression to the Stage 1 scores.

Regression baselines. We compare our CAIRO models to three standard regression baselines: a two-layer MLP with the same architecture trained with squared-error loss (NN-MSE); a random forest built from CART regression trees; and a LightGBM model trained with mean-squared error. This allows a direct comparison between pointwise regression, tree ensembles, and neural ranking-based approaches.

Evaluation Metrics. For ranking quality, we use Spearman’s rank correlation $\hat{\rho}_S$ and Kendall’s $\hat{\tau}$ between y_i and $\hat{y}_i$. Let R_i and R'_i denote the ranks of y_i and $\hat{y}_i$ (with average ranks for ties). The empirical Spearman correlation and Kendall’s $\hat{\tau}$ are

$$\hat{\rho}_S = 1 - \frac{6 \sum_{i=1}^n (R_i - R'_i)^2}{n(n^2 - 1)}, \quad \hat{\tau} = \frac{C - D}{C + D},$$

where C and D are the numbers of concordant and discordant pairs over all $i < j$. Pointwise regression accuracy is summarized by the root mean squared error

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}.$$

We use the same metrics for both the synthetic experiments and the real-data study.

6.1. Synthetic Data-Generating Processes

To isolate the impact of noise topology, we fix the underlying signal structure to be **strictly linear** in the latent space. For all experiments, covariates X and weights w are drawn from $\mathcal{N}(0, I_d)$. We define a shared linear ranking score:

$$\eta = \frac{X^\top w}{\sqrt{d}}.$$

Since η is the sole determinant of the true signal, the optimal ranking is always given by sorting η . We then generate observed responses Y by applying three distinct error mechanisms to this linear backbone:

1. **Normal (Linear, Homoskedastic):** A standard linear regression setting where $Y = \eta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$. This serves as a baseline where MSE is theoretically optimal.
2. **Gamma (Log-Linear, Skewed):** We apply a canonical log-link, setting $\mu = \exp(\eta)$. Responses are drawn from $Y \sim \text{Gamma}(k, \mu/k)$ with shape $k = 2$. Here, variance scales with the mean ($\text{Var}(Y|X) \propto \mu^2$), introducing heteroskedasticity while preserving the linear ordering of the parameters.
3. **Heteroskedastic Tail (Heavy-Tailed):** We again use the mean $\mu = \exp(\eta)$ but introduce multiplicative heavy-tailed noise: $Y = \mu + \varepsilon \cdot \sqrt{\mu}$, where ε is drawn from a Lognormal distribution. This creates extreme outliers that retain the correct rank direction but distort the geometric scale.

Unless stated otherwise, we set $n = 6000$, $d = 10$, and use a 70/30 train-test split.

6.2. Results on Synthetic Data

We evaluate CAIRO against standard regression baselines across the noise regimes described in Section 6.1. Table 2 summarizes the aggregate performance metrics, while Figure 1 provides a granular visual comparison of the trade-offs between regression error (RMSE) and ranking quality (Spearman’s ρ). Full tabular results are presented in Table 2.

Performance in Homoskedastic Regimes. Under standard Normal noise ($\sigma = 1.0$), the NN-MSE baseline corresponds to the maximum likelihood estimator and is theoretically optimal. Consequently, it achieves the lowest RMSE (1.004). In this setting, the challenge for rank-based methods is *fidelity*: recovering the precise linear signal without direct access to residual magnitudes.

We observe that **CAIRO-GiniNet-SoftRank** acts as a high-fidelity proxy, achieving an RMSE of 1.031 and ranking metrics competitive with the baseline ($\rho \approx 0.65$). In contrast, the pairwise **CAIRO-RankNet-GiniW** suffers a slight degradation in both RMSE (1.095) and correlation ($\rho \approx 0.59$). This confirms that in well-behaved regimes, listwise soft-ranking operators preserve fine-grained structural information more effectively than binary, saturating pairwise comparisons.

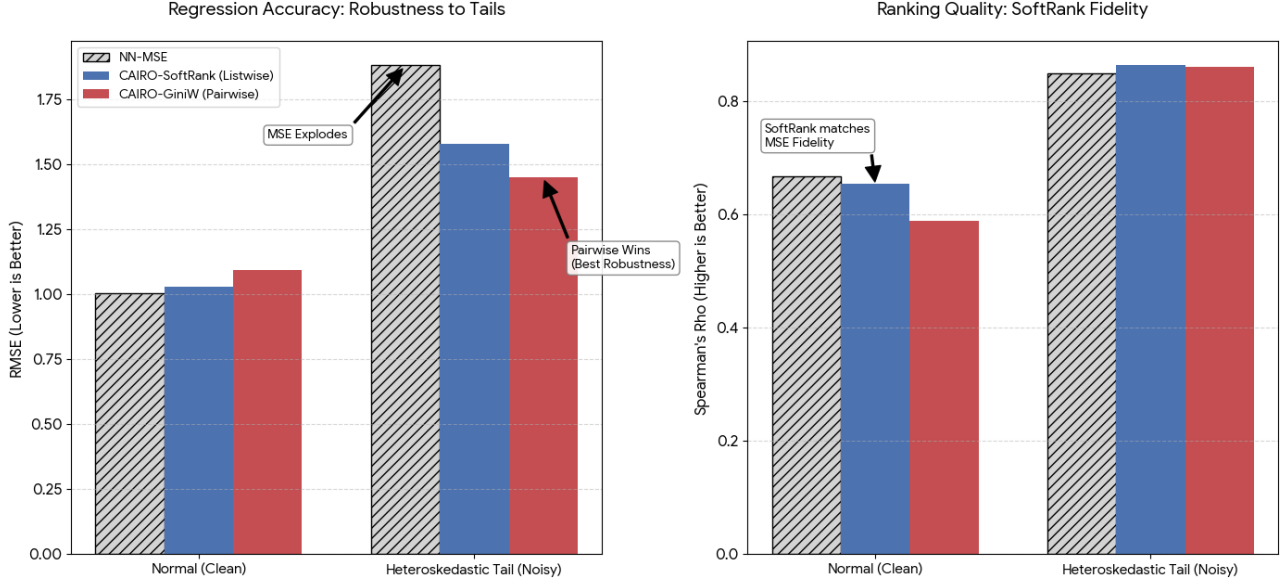


Figure 1. Fidelity vs. Robustness. Comparison of RMSE (left) and Spearman’s ρ (right) across noise regimes. We compare two Gini-oriented objectives: Weighted Pairwise (Red) vs. Listwise SoftRank (Blue). In the Normal regime, CAIRO–SoftRank matches the fidelity of the optimal MSE baseline. In the Heavy-Tailed regime, the baseline explodes due to geometric sensitivity, while the Pairwise Gini (red) demonstrates superior robustness via gradient saturation.

Performance in Heavy-Tailed Regimes. The advantages of the CAIRO framework become decisive in the *Heteroskedastic Tail* regime. Here, the geometric sensitivity of standard regression is exposed: the RMSE of NN–MSE degrades severely to 1.882, as the model over-fits to extreme outliers to minimize quadratic penalties.

As hypothesized, CAIRO demonstrates **topological robustness**. By decoupling order learning from scale calibration, it prevents outliers from distorting the learned representation. Notably, we observe a performance crossover between the ranking surrogates:

- **Sigmoid Saturation as a Shield:** The **CAIRO–RankNet–GiniW** variant achieves the best global performance (RMSE 1.449), outperforming both the baseline and the SoftRank variant (1.580). We attribute this to the bounded influence of the weighted logistic loss. Unlike the quadratic growth of MSE, this loss saturates, effectively “clipping” the gradient contribution of extreme outliers during Stage 1.
- **SoftRank Sensitivity:** While CAIRO–SoftRank still outperforms the uncalibrated baseline (RMSE 1.580), it is less robust than the sigmoid variant. The global soft-sort operator attempts to fit the entire list structure, retaining some sensitivity to the relative position of outliers.

Summary. These results characterize CAIRO as a flexible robust regression framework. For high-fidelity approximation in clean data, SoftRank objectives are preferred. However, for maximum robustness in contaminated environments, the saturating pairwise RankNet objective provides the strongest defense against geometric distortion.

6.3. Real-Data Analysis

Using artificial data is useful for investigating the behaviour of the methods under controlled conditions. However, it is also important to assess their performance on real data sets, to understand how well our CAIRO models generalize in practice.

For this purpose, we use five benchmark regression data sets from the UCI Machine Learning Repository (?): Auto MPG (AUTO), Communities and Crime (CCRI), Computer Hardware (COMP), Concrete Slump Test (CSLU), and Abalone (ABALONE). Their main characteristics are summarized in Table 3. Sample sizes range from $n = 103$ to $n = 4177$, and the number of predictors from $p = 6$ to $p = 102$.

In AUTO, the *car name* predictor is removed because it contains too many distinct categorical levels, and six of the original 398 observations with missing values are discarded, leaving $n = 392$ complete cases. In CCRI, we drop 25 out of the original 128 predictors that each contain at least 58% missing values, resulting in $p = 102$ covariates. For ABALONE, we discard the categorical *Sex* variable and retain only the seven continuous shell measurements as predictors; the response is taken to be age ($\text{Rings} + 1.5$), as in the original UCI description. After these preprocessing steps, there are no missing values in any of the real data sets used in our study. This preprocessing procedure follows the setup of ?.

We validate the transferability of our findings on five UCI regression benchmarks, characterized by varying sample sizes ($n = 103$ to 4177) and feature dimensionalities. Table 1 reports the mean performance across N_{rep} random splits.

Stabilizing Neural Regression. The most immediate observation is the fragility of the standard neural baseline. On datasets such as AUTO, COMP, and CSLU, the NN-MSE model fails to learn a stable predictor, exhibiting variance orders of magnitude larger than other methods. In stark contrast, all CAIRO variants produce stable, high-performance models using the exact same network architecture. This indicates that the "Rank-then-Calibrate" framework acts as a powerful regularizer, enabling neural networks to learn effectively even in small-sample tabular regimes where they typically struggle.

Competitiveness with Tree Ensembles. While Gradient Boosted Trees and Random Forests are generally considered the gold standard for tabular regression, CAIRO proves to be highly competitive, matching or exceeding their performance on three out of five benchmarks:

- **Robustness on COMP:** On the Computer Hardware dataset, which contains significant price outliers, **CAIRO-RankNet-GiniW** achieves an RMSE of **73.7**, substantially outperforming Random Forest (87.8) and LightGBM (94.2). This mirrors our synthetic findings: the weighted pairwise loss provides robustness against large geometric deviations.
- **Small-Data Efficiency on CSLU:** On the Concrete Slump Test ($n = 103$), CAIRO dominates all baselines, with the GiniW variant achieving an RMSE of **3.52** versus 4.45 for Random Forest. This suggests that ranking objectives may be more data-efficient than variance-minimizing objectives when samples are scarce.
- **Ranking Fidelity on ABALONE:** On the larger ABALONE task, CAIRO variants achieve both the highest ranking correlations ($\rho \approx 0.78$) and the lowest RMSE (2.12), slightly edging out the tree ensembles.

Summary. These results confirm that the theoretical properties of CAIRO—specifically the robustness of the pairwise Gini objective—translate to practical gains on real-world data. By decoupling the learning of topology from scale, CAIRO provides a neural regression method that is robust enough to compete with, and often outperform, state-of-the-art tree ensembles.

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A. Appendix

A.1. Tables and Results

Table 1. Real-data performance on five UCI regression tasks. Entries are mean (95% CI) over 5 random 70/30 train–test splits. Best value per dataset/metric is in bold.

Dataset	Model	Spearman	Kendall’s τ	RMSE
AUTO (Auto MPG)				
	CAIRO–GiniNet–SoftRank	0.870 ± 0.022	0.719 ± 0.022	4.244 ± 0.294
	CAIRO–RankNet	0.933 ± 0.011	0.795 ± 0.020	3.188 ± 0.204
	CAIRO–RankNet–GiniW	0.923 ± 0.025	0.787 ± 0.027	3.405 ± 0.455
	NN–MSE	0.050 ± 0.578	0.026 ± 0.434	12.321 ± 4.540
	Random Forest	0.937 ± 0.015	0.798 ± 0.017	2.994 ± 0.239
	LightGBM–MSE	0.935 ± 0.015	0.796 ± 0.019	3.010 ± 0.230
CCRI (Communities and Crime)				
	CAIRO–GiniNet–SoftRank	0.810 ± 0.018	0.640 ± 0.020	0.142 ± 0.002
	CAIRO–RankNet	0.757 ± 0.013	0.586 ± 0.013	0.160 ± 0.005
	CAIRO–RankNet–GiniW	0.762 ± 0.010	0.593 ± 0.011	0.159 ± 0.004
	NN–MSE	0.475 ± 0.371	0.358 ± 0.280	0.190 ± 0.053
	Random Forest	0.817 ± 0.015	0.631 ± 0.016	0.138 ± 0.004
	LightGBM–MSE	0.812 ± 0.012	0.626 ± 0.013	0.138 ± 0.003
COMP (Computer Hardware)				
	CAIRO–GiniNet–SoftRank	0.784 ± 0.042	0.644 ± 0.042	105.055 ± 64.484
	CAIRO–RankNet	0.800 ± 0.072	0.657 ± 0.070	89.822 ± 31.436
	CAIRO–RankNet–GiniW	0.833 ± 0.043	0.688 ± 0.045	73.741 ± 34.310
	NN–MSE	0.392 ± 0.363	0.297 ± 0.294	334.020 ± 142.277
	Random Forest	0.904 ± 0.022	0.753 ± 0.031	87.826 ± 27.136
	LightGBM–MSE	0.808 ± 0.077	0.638 ± 0.079	94.157 ± 27.191
CSLU (Concrete Slump Test)				
	CAIRO–GiniNet–SoftRank	0.769 ± 0.114	0.604 ± 0.106	5.207 ± 0.895
	CAIRO–RankNet	0.822 ± 0.189	0.717 ± 0.145	4.807 ± 2.306
	CAIRO–RankNet–GiniW	0.900 ± 0.013	0.760 ± 0.027	3.522 ± 0.420
	NN–MSE	0.033 ± 0.409	0.038 ± 0.320	23.394 ± 12.779
	Random Forest	0.890 ± 0.030	0.733 ± 0.043	4.445 ± 0.394
	LightGBM–MSE	0.844 ± 0.034	0.682 ± 0.042	4.519 ± 0.251
ABALONE				
	CAIRO–GiniNet–SoftRank	0.631 ± 0.040	0.503 ± 0.034	2.610 ± 0.042
	CAIRO–RankNet	0.781 ± 0.014	0.637 ± 0.013	2.115 ± 0.042
	CAIRO–RankNet–GiniW	0.778 ± 0.015	0.635 ± 0.013	2.116 ± 0.036
	NN–MSE	0.634 ± 0.028	0.481 ± 0.023	3.255 ± 0.102
	Random Forest	0.763 ± 0.019	0.606 ± 0.017	2.172 ± 0.023
	LightGBM–MSE	0.761 ± 0.015	0.603 ± 0.014	2.192 ± 0.045

Table 2. Synthetic-data performance across noise models: mean (95% CI) over 5 repetitions. Best value per scenario/metric is in bold.

Scenario	Model	RMSE	Spearman	Kendall's τ
Normal ($\sigma = 1.0$)				
	CAIRO–RankNet	1.093 ± 0.026	0.594 ± 0.074	0.435 ± 0.059
	CAIRO–RankNet–GiniW	1.095 ± 0.025	0.588 ± 0.074	0.431 ± 0.056
	CAIRO–GiniNet–SoftRank	1.031 ± 0.010	0.654 ± 0.051	0.484 ± 0.041
	NN–MSE	1.004 ± 0.008	0.667 ± 0.045	0.481 ± 0.037
	Random Forest	1.039 ± 0.020	0.642 ± 0.045	0.459 ± 0.037
	LightGBM–MSE	1.033 ± 0.019	0.643 ± 0.056	0.461 ± 0.046
Gamma (shape $k = 2$)				
	CAIRO–RankNet	2.238 ± 0.726	0.701 ± 0.058	0.527 ± 0.050
	CAIRO–RankNet–GiniW	2.082 ± 0.577	0.714 ± 0.058	0.540 ± 0.050
	CAIRO–GiniNet–SoftRank	2.040 ± 0.554	0.730 ± 0.040	0.553 ± 0.037
	NN–MSE	2.125 ± 0.775	0.736 ± 0.040	0.544 ± 0.037
	Random Forest	2.176 ± 0.783	0.698 ± 0.034	0.510 ± 0.030
	LightGBM–MSE	2.112 ± 0.687	0.695 ± 0.037	0.508 ± 0.032
Heteroskedastic tail				
	CAIRO–RankNet	1.423 ± 0.089	0.858 ± 0.037	0.693 ± 0.043
	CAIRO–RankNet–GiniW	1.449 ± 0.095	0.860 ± 0.039	0.696 ± 0.044
	CAIRO–GiniNet–SoftRank	1.580 ± 0.239	0.863 ± 0.033	0.698 ± 0.039
	NN–MSE	1.882 ± 0.513	0.848 ± 0.044	0.667 ± 0.047
	Random Forest	1.873 ± 0.533	0.830 ± 0.026	0.642 ± 0.028
	LightGBM–MSE	1.705 ± 0.470	0.854 ± 0.026	0.672 ± 0.030

Table 3. Characteristics of the real data sets used in our study.

Data set	No. of data points (n)	No. of predictors (p)	No. of unique response values	Source
AUTO	392	7	127	UCI
CCRI	1994	102	98	UCI
COMP	209	6	116	UCI
CSLU	103	7	90	UCI
ABALONE	4177	7	29	UCI

A.2. Proof of Composition Theorem

Proof. Let g be the output from the first stage. Then $g = h(m^*)$ for a strictly increasing function h . We want to characterize the second-stage function f such that

$$f = \arg \min_{t \in \mathcal{M}} \mathbb{E} [(Y - t(g(X)))^2].$$

By adding and subtracting m^* in the objective and expanding the square, we obtain

$$\begin{aligned} \mathbb{E} [(Y - t(g(X)))^2] &= \mathbb{E} [(Y - m^* + m^* - t(g(X)))^2] \\ &= \mathbb{E} [(Y - m^*)^2] + \mathbb{E} [(m^* - t(g(X)))^2] \\ &\quad + 2 \mathbb{E} [(Y - m^*) (m^* - t(g(X)))]. \end{aligned}$$

The first term $\mathbb{E} [(Y - m^*)^2]$ is independent of t , since it depends only on the true regression function m^* . To show that the

cross term is zero, apply the law of iterated expectations (tower property):

$$\begin{aligned}\mathbb{E}_{Y,X}[(Y - m^*)(m^* - t(g(X)))] &= \mathbb{E}_X[\mathbb{E}_{Y|X}[(Y - m^*)(m^* - t(g(X))) | X]] \\ &= \mathbb{E}_X[(m^* - t(g(X))) \mathbb{E}_{Y|X}[Y - m^* | X]] \\ &= 0,\end{aligned}$$

since $\mathbb{E}_{Y|X}[Y - m^* | X] = 0$. Hence the objective simplifies to

$$\arg \min_{t \in \mathcal{M}} \mathbb{E}[(Y - t(g(X)))^2] = \arg \min_{t \in \mathcal{M}} \mathbb{E}[(m^* - t(g(X)))^2].$$

This is minimized when $t(g(X)) = m^*(X)$, i.e.,

$$t(h(m^*(X))) = m^*(X) \Rightarrow t = h^{-1}.$$

The inverse exists since h is strictly increasing, and $h^{-1} \in \mathcal{M}$ because it is non-decreasing as the inverse of a strictly increasing function. Therefore, the optimal second-stage function is $f = h^{-1}$, and

$$f(g(X)) = h^{-1}(h(m^*(X))) = m^*(X).$$

□

A.3. Covariance form of Weighted Pairwise Loss

Lemma A.1. *Let (X, Y) and (X', Y') be i.i.d. with finite second moments and independent across pairs. Then*

$$\mathbb{E}[(Y - Y')(X - X')] = 2 \text{Cov}(Y, X).$$

Proof.

$$\begin{aligned}\mathbb{E}[(Y - Y')(X - X')] &= \mathbb{E}[YX] - \mathbb{E}[YX'] - \mathbb{E}[Y'X] + \mathbb{E}[Y'X'] \\ &= \mathbb{E}[YX] - \mathbb{E}[Y]\mathbb{E}[X'] - \mathbb{E}[Y']\mathbb{E}[X] + \mathbb{E}[Y'X'] \\ &= \mathbb{E}[YX] - \mathbb{E}[Y]\mathbb{E}[X] - \mathbb{E}[Y]\mathbb{E}[X] + \mathbb{E}[YX] \\ &= 2(\mathbb{E}[YX] - \mathbb{E}[Y]\mathbb{E}[X]) = 2 \text{Cov}(Y, X).\end{aligned}$$

In the second line, independence across pairs gives $\mathbb{E}[YX'] = \mathbb{E}[Y]\mathbb{E}[X']$ and $\mathbb{E}[Y'X] = \mathbb{E}[Y']\mathbb{E}[X]$. In the third line, i.i.d. across pairs yields $\mathbb{E}[Y'X'] = \mathbb{E}[YX]$, $\mathbb{E}[X'] = \mathbb{E}[X]$, and $\mathbb{E}[Y'] = \mathbb{E}[Y]$. □

Lemma A.2. *Let X and X' be i.i.d. real random variables with continuous cdf F . With $\text{sgn}(t) = \mathbf{1}\{t > 0\} - \mathbf{1}\{t < 0\}$, we have*

$$\mathbb{E}[\text{sgn}(X - X') | X] = 2F(X) - 1 \quad a.s.$$

Proof. Use the decomposition: $\text{sgn}(X - X') = \mathbf{1}\{X > X'\} - \mathbf{1}\{X < X'\}$. Then:

$$\begin{aligned}\mathbb{E}[\text{sgn}(X - X') | X] &= \mathbb{E}[\mathbf{1}\{X > X'\} - \mathbf{1}\{X < X'\} | X] \\ &= \mathbb{P}(X' < X | X) - \mathbb{P}(X' > X | X) \\ &= F(X) - (1 - F(X)) \\ &= 2F(X) - 1\end{aligned}$$

□

Proposition A.3. *Let (X, Y) and (X', Y') be i.i.d. with finite second moments and independent across pairs. Let $f : \mathbb{R}^q \rightarrow \mathbb{R}$ be measurable and assume $f(X)$ has a continuous cdf F_f (so $\mathbb{P}(f(X) = f(X')) = 0$). Then*

$$\text{Cov}(Y, F_g(g(X))) = \frac{1}{2} \mathbb{E}[(Y - Y') \mathbf{1}\{g(X) > g(X')\}].$$

Proof. Let (X', Y') be an i.i.d. copy of (X, Y) , and assume $\mathbb{P}(g(X) = g(X')) = 0$ so that ties occur with probability zero.

$$\begin{aligned}
 2 \cdot \text{Cov}(Y, F_g(g(X))) &= 2 \left(\mathbb{E}[Y F_g(g(X))] - \mathbb{E}[Y] \mathbb{E}[F_g(g(X))] \right) \\
 &= 2 \left(\mathbb{E}[Y F_g(g(X))] - \frac{1}{2} \mathbb{E}[Y] \right) \\
 &= 2 \mathbb{E}[Y F_g(g(X))] - \mathbb{E}[Y] \\
 &= 2 \mathbb{E}[Y \mathbb{E}[\mathbb{1}\{g(X') \leq g(X)\} \mid X]] - \mathbb{E}[Y] \\
 &= 2 \mathbb{E}[Y \mathbb{1}\{g(X') \leq g(X)\}] - \mathbb{E}[Y].
 \end{aligned} \tag{13}$$

By symmetry of the i.i.d. pairs (X, Y) and (X', Y') ,

$$2 \mathbb{E}[Y \mathbb{1}\{g(X') \leq g(X)\}] = \mathbb{E}[Y \mathbb{1}\{g(X') \leq g(X)\}] + \mathbb{E}[Y' \mathbb{1}\{g(X) \leq g(X')\}]. \tag{14}$$

Using $\mathbb{P}(g(X) = g(X')) = 0$, we may replace $\leq$ by $<$ in the indicators. Combining (13) and (14), we obtain

$$\begin{aligned}
 2 \cdot \text{Cov}(Y, F_g(g(X))) &= \mathbb{E}[Y \mathbb{1}\{g(X') < g(X)\}] + \mathbb{E}[Y' \mathbb{1}\{g(X) < g(X')\}] - \mathbb{E}[Y] \\
 &= \mathbb{E}[Y \mathbb{1}\{g(X') < g(X)\}] + \mathbb{E}[Y' \mathbb{1}\{g(X) < g(X')\}] \\
 &\quad - \mathbb{E}[Y (\mathbb{1}\{g(X') < g(X)\} + \mathbb{1}\{g(X) < g(X')\})] \\
 &= \mathbb{E}[Y' \mathbb{1}\{g(X) < g(X')\}] - \mathbb{E}[Y \mathbb{1}\{g(X) < g(X')\}] \\
 &= \mathbb{E}[(Y' - Y) \mathbb{1}\{g(X) < g(X')\}] \\
 &= \mathbb{E}[(Y - Y') \mathbb{1}\{g(X) > g(X')\}].
 \end{aligned} \tag{15}$$

This proves

$$2 \cdot \text{Cov}(Y, F_g(g(X))) = \mathbb{E}[(Y - Y') \mathbb{1}\{g(X) > g(X')\}],$$

which is the desired identity. $\square$

Proposition A.4. Let (X, Y) and (X', Y') be i.i.d. copies with $\mathbb{P}(g(X) = g(X')) = 0$, and define the gap-weighted misranking risk

$$\mathcal{L}_{\text{gap}}(g) := \mathbb{E}[|Y - Y'| \mathbb{1}\{(Y - Y')(g(X) - g(X')) < 0\}].$$

Then

$$\mathcal{L}_{\text{gap}}(g) = \frac{1}{2} \mathbb{E}[|Y - Y'|] - 2 \text{Cov}(Y, F_g(g(X))).$$

Proof. Let

$$\begin{aligned}
 T &:= \mathbb{E}[|Y - Y'|], \\
 C &:= \mathbb{E}[|Y - Y'| (\mathbb{1}\{Y > Y', g(X) > g(X')\} + \mathbb{1}\{Y < Y', g(X) < g(X')\})], \\
 D &:= \mathbb{E}[|Y - Y'| (\mathbb{1}\{Y > Y', g(X) < g(X')\} + \mathbb{1}\{Y < Y', g(X) > g(X')\})].
 \end{aligned}$$

These are the total mass, the gap-weighted concordant mass, and the gap-weighted discordant mass, respectively. Clearly $T = C + D$ and $\mathcal{L}_{\text{gap}}(g) = D$.

Using the covariance-pairwise identity established earlier,

$$2 \text{Cov}(Y, F_g(g(X))) = \mathbb{E}[(Y - Y') \mathbb{1}\{g(X) > g(X')\}] \tag{16}$$

$$= \mathbb{E}[|Y - Y'| \text{sgn}(Y - Y') \frac{1}{2} (\text{sgn}(g(X) - g(X')) + 1)] \tag{17}$$

$$= \frac{1}{2} \mathbb{E}[|Y - Y'| \text{sgn}(Y - Y') \text{sgn}(g(X) - g(X'))] \tag{18}$$

$$= \frac{1}{2} (C - D). \tag{19}$$

From $T = C + D$ and $C - D = 4 \text{Cov}(Y, F_g(g(X)))$ we obtain

$$D = \frac{1}{2} T - 2 \text{Cov}(Y, F_g(g(X))),$$

so

$$\mathcal{L}_{\text{gap}}(g) = D = \frac{1}{2} \mathbb{E}[|Y - Y'|] - 2 \text{Cov}(Y, F_g(g(X))),$$

which proves the claim. $\square$

Theorem A.5. Consider the pair (X, Y) . Assume $\mathbb{E}|Y| < \infty$. For a measurable score $f : \mathcal{X} \rightarrow \mathbb{R}$, let F_f be the cdf of $f(X)$ and suppose F_f is continuous. Consider the loss

$$\mathcal{L}(f) = -\text{Cov}(Y, F_f(f(X))).$$

Then the minimizers of $\mathcal{L}$ are exactly the functions that induce the same strict ordering as the regression function; In particular, any $f = \phi \circ m^*$ with strictly increasing ϕ is a minimizer, and every minimizer agrees with such an f in its induced order almost surely.

Moreover,

$$\max_f \text{Cov}(Y, F_f(f(X))) = \frac{1}{4} \mathbb{E}|m^*(X) - m^*(X')|.$$

Proof. By the proved identity A.3,

$$\text{Cov}(Y, F_f(f(X))) = \frac{1}{2} \mathbb{E}[(Y - Y') \mathbf{1}\{f(X) > f(X')\}]. \quad (*)$$

Taking conditional expectation given (X, X') ,

$$\mathbb{E}[Y - Y' \mid X, X'] = m^*(X) - m^*(X'),$$

so

$$\text{Cov}(Y, F_f(f(X))) = \frac{1}{2} \mathbb{E}[(m^*(X) - m^*(X')) \mathbf{1}\{f(X) > f(X')\}]. \quad (1)$$

Upper bound. For any $a \in \mathbb{R}$ and $s \in \{0, 1\}$, $as \leq a_+$ with equality iff $s = \mathbf{1}\{a > 0\}$. Apply this pointwise to $a = m^*(X) - m^*(X')$ and $s = \mathbf{1}\{f(X) > f(X')\}$:

$$(m^*(X) - m^*(X')) \mathbf{1}\{f(X) > f(X')\} \leq (m^*(X) - m^*(X'))_+.$$

Taking expectations and using the identity $\mathbb{E}[(A - A')_+] = \frac{1}{2} \mathbb{E}|A - A'|$ for i.i.d. A, A' ,

$$\text{Cov}(Y, F_f(f(X))) \leq \frac{1}{2} \mathbb{E}[(m^*(X) - m^*(X'))_+] = \frac{1}{4} \mathbb{E}|m^*(X) - m^*(X')|. \quad (2)$$

Achievability and characterization. Equality in the pointwise bound holds iff

$$\mathbf{1}\{f(X) > f(X')\} = \mathbf{1}\{m^*(X) > m^*(X')\} \quad \text{a.s.} \quad (3)$$

Thus any f that induces the same strict order as m^* attains equality in (2) and achieves the maximum in the theorem. One way is $f = \phi \circ m^*$ with strictly increasing ϕ , chosen so that $f(X)$ has a continuous cdf. Conversely, if f is optimal then equality must hold in (2) almost surely, forcing (3). This order-equivalence is exactly the stated characterization. The optimal value is then the right-hand side of (2). $\square$

A.4. Alternative expression for the empirical weighted misranking loss

Proposition A.6. Assume $\{(x_i, y_i)\}_{i=1}^n$ is such that there are no ties in $\{y_i\}$ and in $\{g(x_i)\}$, and $w_{ij} = w_{ji} \geq 0$ for all $i \neq j$. Then the empirical loss defined in (4) satisfies

$$\ell_w(g) = \frac{1}{n(n-1)} \sum_{i \neq j} w_{ij} \mathbf{1}\{y_i > y_j\} \mathbf{1}\{g(x_i) < g(x_j)\}.$$

Proof. Because there are no ties, the event $(y_i - y_j)(g(x_i) - g(x_j)) \leq 0$ is equivalent to $(y_i - y_j)(g(x_i) - g(x_j)) < 0$, and this occurs if and only if exactly one of the two comparisons is reversed:

$$\mathbf{1}\{(y_i - y_j)(g(x_i) - g(x_j)) < 0\} = \mathbf{1}\{y_i > y_j, g(x_i) < g(x_j)\} + \mathbf{1}\{y_i < y_j, g(x_i) > g(x_j)\}.$$

Plugging this into (4) yields

$$\ell_w(g) = \frac{1}{n(n-1)} \sum_{1 \leq i < j \leq n} w_{ij} \left[\mathbb{1}\{y_i > y_j, g(x_i) < g(x_j)\} + \mathbb{1}\{y_i < y_j, g(x_i) > g(x_j)\} \right].$$

Now consider the ordered-pair sum

$$S := \sum_{i \neq j} w_{ij} \mathbb{1}\{y_i > y_j\} \mathbb{1}\{g(x_i) < g(x_j)\}.$$

We can rewrite S by grouping terms over unordered pairs $\{i, j\}$ with $i < j$:

$$S = \sum_{1 \leq i < j \leq n} \left(w_{ij} \mathbb{1}\{y_i > y_j, g(x_i) < g(x_j)\} + w_{ji} \mathbb{1}\{y_j > y_i, g(x_j) < g(x_i)\} \right).$$

By symmetry $w_{ji} = w_{ij}$, and since $y_j > y_i$ and $g(x_j) < g(x_i)$ are exactly the negations of $y_i > y_j$ and $g(x_i) < g(x_j)$, we have

$$\mathbb{1}\{y_j > y_i, g(x_j) < g(x_i)\} = \mathbb{1}\{y_i < y_j, g(x_i) > g(x_j)\}.$$

Therefore

$$S = \sum_{1 \leq i < j \leq n} w_{ij} \left[\mathbb{1}\{y_i > y_j, g(x_i) < g(x_j)\} + \mathbb{1}\{y_i < y_j, g(x_i) > g(x_j)\} \right].$$

Comparing this with the expression for $\ell_w(g)$ above, we obtain

$$\ell_w(g) = \frac{1}{n(n-1)} S = \frac{1}{n(n-1)} \sum_{i \neq j} w_{ij} \mathbb{1}\{y_i > y_j\} \mathbb{1}\{g(x_i) < g(x_j)\},$$

which is exactly the claimed identity. □